

Sorting: Comparator

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Comparators are used to compare two objects. In this challenge, you'll create a comparator and use it to sort an array. The `Player` class is provided in the editor below. It has two fields:

1. **`name`**: a string.
2. **`score`**: an integer.

Given an array of n `Player` objects, write a comparator that sorts them in order of decreasing score. If **2** or more players have the same score, sort those players alphabetically ascending by name. To do this, you must create a `Checker` class that implements the `Comparator` interface, then write an `int compare(Player a, Player b)` method implementing the `Comparator.compare(T o1, T o2)` method. In short, when sorting in ascending order, a comparator function returns -1 if $a < b$, 0 if $a = b$, and 1 if $a > b$.

Declare a `Checker` class that implements the comparator method as described. It should sort first descending by score, then ascending by name. The code stub reads the input, creates a list of `Player` objects, uses your method to sort the data, and prints it out properly.

Example

$n = 3$ `data = [[Smith, 20], [Jones, 15], [Jones, 20]]`

Sort the list as `data_sorted = [[Jones, 20], [Smith, 20], [Jones, 15]]`. Sort first descending by score, then ascending by name.

Input Format

The first line contains an integer, n , the number of players.

Each of the next n lines contains a player's **`name`** and **`score`**, a string and an integer.

Constraints

- $0 \leq \text{score} \leq 1000$
- Two or more players can have the same name.
- Player names consist of lowercase English alphabetic letters.

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Difficulty [Medium](#)
Max Score **35**
Submitted By [150401](#)

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Change Theme Language Java 7

```
1 import java.util.*;
2
3 class Player {
4     String name;
5     int score;
6
7     Player(String name, int score) {
8         this.name = name;
9         this.score = score;
10    }
11 }
12
13 class Checker implements Comparator<Player> {
14
15     public int compare(Player a, Player b) {
16         if (a.score > b.score) {
17             return -1;
18         }
19         if (a.score < b.score) {
20             return 1;
21         }
22         return a.name.compareTo(b.name);
23     }
24 }
25
26
27
28 public class Solution {
29
30     public static void main(String[] args) {
31         Scanner scan = new Scanner(System.in);
32         int n = scan.nextInt();
33
34         Player[] player = new Player[n];
35         Checker checker = new Checker();
36
37         for(int i = 0; i < n; i++){
38             player[i] = new Player(scan.next(), scan.nextInt());
39         }
40     }
41 }
```

Line: 25 Col: 1

☐ Test against custom input

Submit Code

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 2

1	5
2	amy 100
3	david 100
4	heraldo 50
5	aakansha 75
6	aleksa 150

1	aleksa 150
2	amy 100
3	david 100